

Ethan Mader

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EDUCATION

Computer Science Ph.D., Purdue | Aug 2024 - Present | West Lafayette, IN

GPA: 3.94/4.0; focus on theoretical computer science, advised by Kent Quanrud

Computer Science B.S., Mathematics B.S., UCSB | Sep 2020 - June 2024 | Goleta, CA

GPA: 3.93/4.0; **SAT:** Math 800; **SAT Math II:** 800; Theta Tau Professional Engineering Fraternity

QUANT SKILLS: Python (Numpy, Pandas, CVX, Scikit-Learn), C++, SQL, Convex Optimization, Linear Algebra, Graph Algorithms, Randomized Algorithms, Backtesting, Statistical Modeling, LaTeX

AWARDS & HONORS

- **Presidential Excellence PhD Award:** \$40,000 over 4 years given to the top 1.4% of incoming PhD students at Purdue in 2024
- **Herbold Scholarship:** Selected to receive \$10,000 out of 200+ CS PhD students at Purdue
- **UCSB Dean's Honors:** Awarded for 11 of 12 academic quarters
- UCSB *magna cum laude* in College of L.S., *cum laude* in CoE
- 82nd Putnam exam: Scored in the top half of the country

RESEARCH

Research Assistant, Purdue | Sep 2024 - Present | West Lafayette, IN

- Improved upon randomized techniques to make runtime improvements for the classic edit distance problem under the relaxation to a 3-approximation
- Summarized state-of-the-art graph decomposition techniques to optimize the efficiency of flow-based algorithms on large networks

Undergrad Researcher, UCSB Programming Languages Lab | Jan - June 2023 | Goleta, CA

- Developed novel techniques for algorithmically generating mathematical proofs via code, enhancing formal verification techniques
- Research findings led to the translation of a theorem-proving package from Agda to Coq for use in circuit verification

TEACHING

Math and CS Tutor, Wyzant | June 2023 - Sep 2024 | CA

- Remote instruction on competition math and C++ fundamentals, data structures, and algorithms
- 200+ hours and perfect 5-star rating across 30 reviews, with one-time and long-term clients

PROJECTS

Matrix Tensor Factorization

- Used alternating least squares with graduated regularization and symmetry constraints in Python to obtain low rank matrix tensor factorizations
- Recovered Strassen's algorithm, and found new exact factorizations for multiplying 3 by 3 matrices

INTERESTS: Jane Street monthly puzzles, bullet chess (top 2% on chess.com), rock climbing, tennis, hiking